

Selenium XPath Practical

Source code and output screenshots

1. Source Code (TC01 - TC20)

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.webdriver.support.ui import Select
import time

driver = webdriver.Chrome()

driver.maximize_window()

# TC01 - Open student registration page get()

driver.get("https://www.selenium.dev/selenium/web/web-form.html")

time.sleep(3)

# TC02 - Locate username Attribute XPath

username = driver.find_element(
    By.XPATH,
    "//input[@name='my-text']"
)

username.clear()
username.send_keys("kavin")

time.sleep(2)

# TC03 - Enter password Attribute XPath

password = driver.find_element(
    By.XPATH,
    "//input[@name='my-password']"
)

password.send_keys("pass1234")

time.sleep(2)

# TC04 - Locate Submit text()

submit_button = driver.find_element(
    By.XPATH,
    "//button[text()='Submit']"
)

print("Submit button:", submit_button.text)

time.sleep(2)

# TC05 - Locate textbox dynamically contains()

textarea = driver.find_element(
    By.XPATH,
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        "//textarea[contains(@name,'textarea'))]"
    )

    textarea.send_keys("This is Selenium XPath testing")

    time.sleep(2)

# TC06 - Locate element with prefix starts-with()

datalist = driver.find_element(
    By.XPATH,
    "//input[starts-with(@name,'my-data'))]"
)

datalist.send_keys("One")

time.sleep(2)

# TC07 - Find input using two attributes and

password = driver.find_element(
    By.XPATH,
    "//input[@name='my-password' and @type='password']"
)

print("Password field found using two attributes")

time.sleep(2)

# TC08 - Find element using alternatives or

text_or_password = driver.find_element(
    By.XPATH,
    "//input[@name='my-text' or @name='my-password']"
)

print("Element found using OR")

time.sleep(2)

# TC09 - Find parent form parent

parent = driver.find_element(
    By.XPATH,
    "//input[@name='my-text']/parent::*"
)

print("Parent tag:", parent.tag_name)

time.sleep(2)

# TC10 - Find form from input ancestor

form = driver.find_element(
    By.XPATH,
    "//input[@name='my-text']/ancestor::form"
)

print("Form found using ancestor:", form.tag_name)

time.sleep(2)

# TC11 - Find child inputs child
child_inputs = driver.find_elements(

```

```

        By.XPATH,
        "//form/descendant::input"
    )

print("Child input fields:", len(child_inputs))

time.sleep(2)

# TC12 - Find next element following

following_element = driver.find_element(
    By.XPATH,
    "//input[@name='my-text']/following::input[1]"
)

print(
    "Following element:",
    following_element.get_attribute("name")
)

time.sleep(2)

# TC13 - Find checkbox Attribute + XPath

checkbox = driver.find_element(
    By.XPATH,
    "//input[@name='my-check']"
)

if not checkbox.is_selected():
    checkbox.click()

print("Checkbox selected")

time.sleep(2)

# TC14 - Find radio button Attribute + XPath

radio = driver.find_element(
    By.XPATH,
    "//input[@name='my-radio']"
)

if not radio.is_selected():
    radio.click()

print("Radio button selected")

time.sleep(2)

# TC15 - Select dropdown XPath + Select

dropdown = Select(
    driver.find_element(
        By.XPATH,
        "//select[@name='my-select']"
    )
)

dropdown.select_by_visible_text("Two")

print("Dropdown value selected: Two")

time.sleep(2)
# TC16 - Find second textbox XPath index
second_textbox = driver.find_element(

```

```

        By.XPATH,
        "(//input)[2]"
    )

print(
    "Second input:",
    second_textbox.get_attribute("name")
)

time.sleep(2)

# TC17 - Verify submitted message text()

submit_button = WebDriverWait(driver, 10).until(
    EC.element_to_be_clickable(
        (By.XPATH, "//button[text()='Submit']")
    )
)

submit_button.click()

time.sleep(3)

success_message = WebDriverWait(driver, 10).until(
    EC.visibility_of_element_located(
        (By.XPATH, "//*[contains(text(),'Received')]")
    )
)

print("Submitted message:", success_message.text)

time.sleep(2)

# TC18 - Find all input fields find_elements()

driver.back()

time.sleep(2)

all_inputs = driver.find_elements(
    By.XPATH,
    "//input"
)

print("Total input fields:", len(all_inputs))

time.sleep(2)

# TC19 - Find dynamic element contains()

dynamic_element = driver.find_element(
    By.XPATH,
    "//input[contains(@name, 'my-')]"
)

print(
    "Dynamic element:",
    dynamic_element.get_attribute("name")
)

time.sleep(2)

# TC20 - Complete registration automation Multiple XPath concepts

username = driver.find_element(
    By.XPATH,

```

```

        "//input[@name='my-text']"
    )

    username.clear()
    username.send_keys("DK")

    time.sleep(1)

    password = driver.find_element(
        By.XPATH,
        "//input[@name='my-password']"
    )

    password.clear()
    password.send_keys("pass1234")

    time.sleep(1)

    textarea = driver.find_element(
        By.XPATH,
        "//textarea[contains(@name,'textarea')]"
    )

    textarea.clear()
    textarea.send_keys("Complete XPath Automation Test")

    time.sleep(1)

    dropdown = Select(
        driver.find_element(
            By.XPATH,
            "//select[@name='my-select']"
        )
    )

    dropdown.select_by_visible_text("Three")

    time.sleep(1)

    checkbox = driver.find_element(
        By.XPATH,
        "//input[@name='my-check']"
    )

    if not checkbox.is_selected():
        checkbox.click()

    time.sleep(1)

    radio = driver.find_element(
        By.XPATH,
        "//input[@name='my-radio']"
    )

    if not radio.is_selected():
        radio.click()

    time.sleep(2)

    submit_button = WebDriverWait(driver, 10).until(
        EC.element_to_be_clickable(
            (By.XPATH, "//button[text()='Submit']")
        )
    )

```

```
submit_button.click()

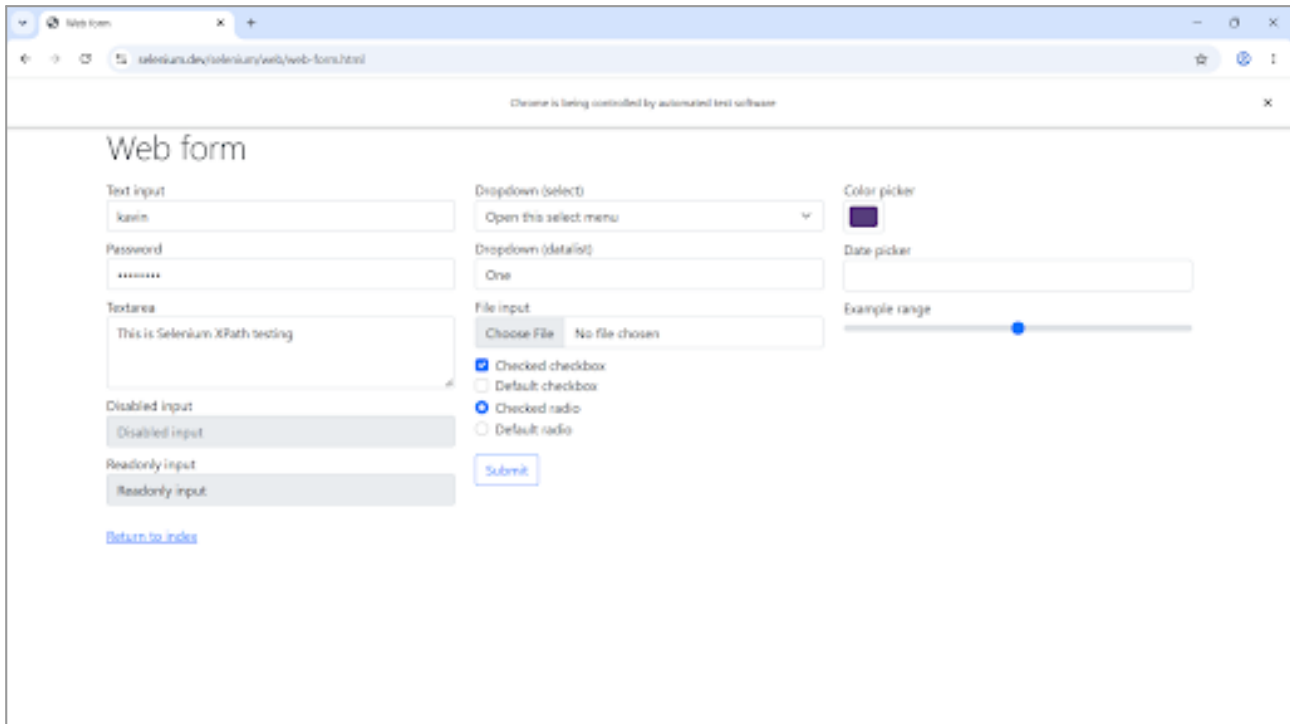
time.sleep(3)

print("Complete registration automation finished")

input("Press Enter to close browser...")

driver.quit()
```

2. Output

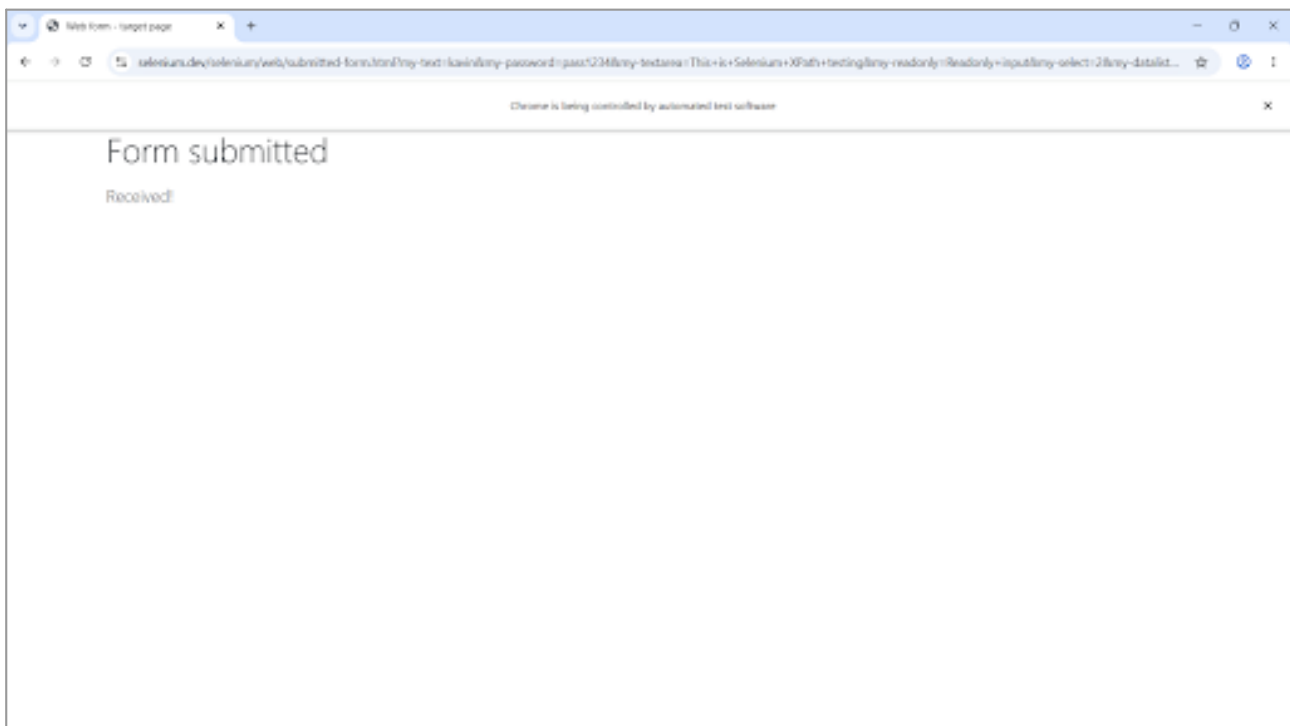


The screenshot shows a web browser window with the address bar displaying 'selenium.dev/selenium/web/web-form.html'. The page title is 'Web form'. The form contains several input fields and controls:

- Text input:** Contains the text 'kavin'.
- Password:** Contains masked text '*****'.
- Textarea:** Contains the text 'This is Selenium XPath testing'.
- Disabled input:** A disabled text input field with the text 'Disabled input'.
- Readonly input:** A readonly text input field with the text 'Readonly input'.
- Dropdown (select):** A dropdown menu with the text 'Open this select menu' and a downward arrow.
- Dropdown (datalist):** A dropdown menu with the text 'One'.
- File input:** A file input field with the text 'Choose File' and 'No file chosen'.
- Checked checkbox:** A checked checkbox with the text 'Checked checkbox'.
- Default checkbox:** An unchecked checkbox with the text 'Default checkbox'.
- Checked radio:** A checked radio button with the text 'Checked radio'.
- Default radio:** An unchecked radio button with the text 'Default radio'.

A 'Submit' button is located at the bottom right of the form. A 'Return to index' link is located at the bottom left of the form.

Fig 1: Web form filled by the script (username, password, textarea, dropdown, checkbox, radio)



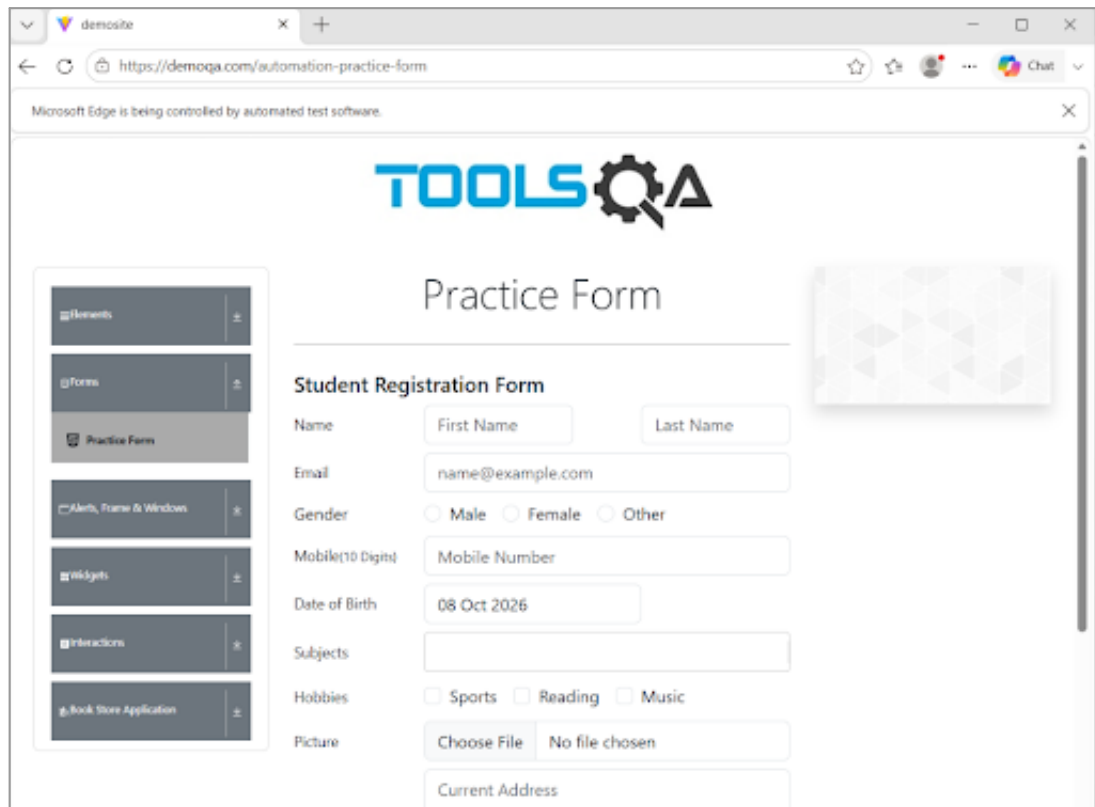
The screenshot shows a web browser window with the address bar displaying 'selenium.dev/selenium/web/submitted-form.html?my-text=kavin&my-password=pass1234&my-textarea=This+is+Selenium+XPath+testing&my-readonly=Readonly+input&my-select=2&my-datalist=One'. The page title is 'Form submitted'. The page content shows the text 'Received!'.

Fig 2: Form submitted page - "Received!" message (TC17)

```
Checkbox selected  
Radio button selected  
Dropdown value selected: Two  
Second input: my-password  
Created TensorFlow Lite XNNPACK delegate for CPU.  
Submitted message: Received!  
Total input fields: 14
```

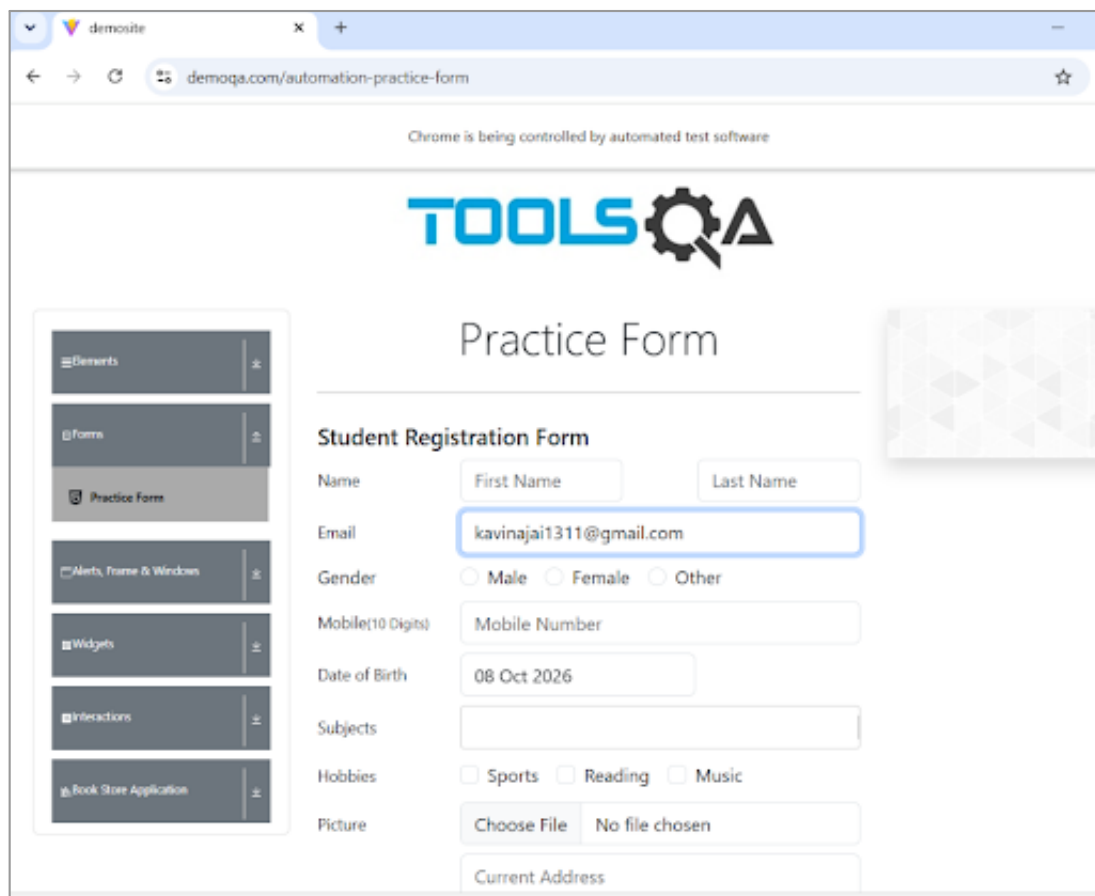
Fig 3: Console output of the script

3. Practice Form Screenshots (demoqa.com)



The screenshot shows the 'Practice Form' page in Microsoft Edge. The browser's address bar displays 'https://demoqa.com/automation-practice-form'. A notification at the top states 'Microsoft Edge is being controlled by automated test software.' The page features the 'TOOLSQA' logo and a sidebar with navigation links: Elements, Forms, Practice Form (selected), Alerts, Frame & Windows, Widgets, Interactions, and Book Store Application. The 'Student Registration Form' contains the following fields: Name (First Name, Last Name), Email (name@example.com), Gender (Male, Female, Other), Mobile(10 Digits) (Mobile Number), Date of Birth (08 Oct 2026), Subjects, Hobbies (Sports, Reading, Music), Picture (Choose File, No file chosen), and Current Address.

Fig 4: Student Registration Form opened in Edge



The screenshot shows the 'Practice Form' page in Chrome. The browser's address bar displays 'demoqa.com/automation-practice-form'. A notification at the top states 'Chrome is being controlled by automated test software.' The page layout is identical to Fig 4. In this version, the 'Email' field is filled with 'kavinajai1311@gmail.com' and is highlighted with a blue border.

Fig 5: Email field filled

The screenshot shows a web browser window with the URL <https://demoqa.com/automation-practice-form>. The browser is Microsoft Edge, and a message at the top states "Microsoft Edge is being controlled by automated test software." The form is titled "Practice Form" and includes a sidebar with navigation links: "Practice Form", "Alerts, Frame & Windows", "Widgets", "Interactions", and "Book Store Application". The form fields are as follows:

- Name:** "First Name" (red border with error icon) and "Last Name" (red border with error icon).
- Email:** "name@example.com" (green checkmark).
- Gender:** Radio buttons for "Male", "Female", and "Other" (all unselected).
- Mobile(10 Digits):** "Mobile Number" (red border with error icon).
- Date of Birth:** "08 Oct 2026" (green checkmark).
- Subjects:** Empty text input.
- Hobbies:** Checkboxes for "Sports", "Reading", and "Music" (all unchecked).
- Picture:** "Choose File" button and "No file chosen" text (green checkmark).
- Current Address:** Empty text input (green checkmark).
- State and City:** "Select State" and "Select City" dropdown menus.
- Submit:** Blue button.

Fig 6: Form validation on empty required fields

The screenshot shows the same web browser window, but the "First Name" field is now filled with the text "Kavin". The browser is Chrome, and a message at the top states "Chrome is being controlled by automated test software." The form is titled "Practice Form" and includes a sidebar with navigation links: "Elements", "Forms", "Practice Form", "Alerts, Frame & Windows", "Widgets", "Interactions", and "Book Store Application". The form fields are as follows:

- Name:** "Kavin" (blue border) and "Last Name" (empty).
- Email:** "name@example.com" (green checkmark).
- Gender:** Radio buttons for "Male", "Female", and "Other" (all unselected).
- Mobile(10 Digits):** "Mobile Number" (empty).
- Date of Birth:** "08 Oct 2026" (green checkmark).
- Subjects:** Empty text input.
- Hobbies:** Checkboxes for "Sports", "Reading", and "Music" (all unchecked).
- Picture:** "Choose File" button and "No file chosen" text (green checkmark).
- Current Address:** Empty text input (green checkmark).
- State and City:** "Select State" and "Select City" dropdown menus.
- Submit:** Blue button.

At the bottom of the page, there is a footer: "© 2013-2026 TOOLSQA.COM | ALL RIGHTS RESERVED."

Fig 7: First Name field filled

GitHub Link: <https://github.com/Kavin1311/08TH-OCT>